Data Structures & Algorithms Laboratory Final Report

Master of Computer Applications — IIPS DAVV, Indore

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1. Course & Laboratory Overview

This comprehensive laboratory report documents the design, rigorous implementation, testing, and mathematical analysis of 7 academic units covering fundamental and advanced data structures and algorithms (IC-307D), supplemented by four End-Semester Capstone Projects.

2. Summary of Implemented Units

Unit	Key Implementations	CO	Status
Unit 1: Arrays	1D/2D arrays, Sparse Matrix Triple Format, Matrix Calculator	CO1	Verified & Tested
Unit 2: Sorting	8 Sorts, Empirical Benchmarking, Student Records, Search Engine	CO2	Verified & Tested
Unit 3: Lists	Singly, Doubly, Circular Lists, Polynomial Addition, Text Editor	CO1	Verified & Tested
Unit 4: Stacks	Static/Dynamic Stacks, Infix/Postfix/Prefix, Evaluator, Undo/Redo	CO1	Verified & Tested
Unit 5: Queues	Circular Ring Buffer, Deque, Priority Queues, OS CPU Scheduler	CO1	Verified & Tested
Unit 6: Trees	Traversals, BST, AVL Rotations, Threaded BST, File System, Trie	CO3	Verified & Tested
Unit 7: Graphs	Adj Matrix/List, BFS/DFS, Dijkstra, Social Network, Maze, DAG	CO3	Verified & Tested

3. Capstone Projects Suite

The repository features complete end-to-end architectures for (1) In-Memory Database Engine with secondary BST indexing; (2) Hospital Management System integrating triage min-heaps and contact tracing graphs; (3) E-Commerce Backend Simulator with BST catalog, cart undo stack, and order queue; and (4) Grid Pathfinding Visualizer comparing BFS and Dijkstra.

4. Verification & Continuous Integration

All modules are accompanied by unit test suites executed via python unittest discover with 100% pass rates across Python 3.10 through 3.13.